DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS
PH1020 Physics II
Lecture Note-2

Multipole Expansion of Potential Date: 1 Feb 2025

1. Multipole Expansion of Potential

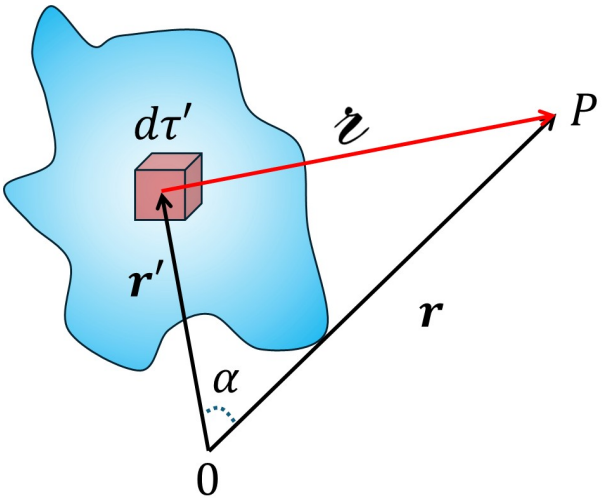


Figure 1: Potential at a point P with position vector $\boldsymbol{r}$ due to localized charge distribution situated around $\boldsymbol{r}'$. Here α is the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$. The separation vector $\boldsymbol{\imath} = \boldsymbol{r} - \boldsymbol{r}'$.

There are often situations that arise where an *observer* is far away from a localized charge distribution $\rho(\boldsymbol{r}')$ and wants to know what the potential $V(\boldsymbol{r})$ (or the electric field) far from the localized charge distribution. The potential at $\boldsymbol{r}$ is given by

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{1}{\imath}\rho(\mathbf{r}')\,d\tau' \tag{1}$$

We can carry out the expansion of the potential and arrive at Griffiths equation (3.96) below

$$V(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r}\int \rho(\boldsymbol{r}')\,d\tau' + \frac{1}{r^2}\int r'\cos\alpha\rho(\boldsymbol{r}')\,d\tau' + \frac{1}{r^3}\int (r')^2\left(\frac{3}{2}\cos^2\alpha - \frac{1}{2}\right)\rho(\boldsymbol{r}')\,d\tau' + \cdots\right] \tag{2}$$

This is called the **multipole expansion of V in powers of $\frac{1}{r}$**. Note each of the terms in the multipole expansion Eq. 2, can be separated into two pieces: one that describes the source (which does not depend on $\boldsymbol{r}$), and one that describes how that source produces a potential (which does depend on $\boldsymbol{r}$).

Lets discuss each term of Eq. 2 in detail below.

1.1 The Monopole term:

The first term ($n = 0$) is the monopole contribution (it goes like $\frac{1}{r}$);

$$V_{\text{mono}}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r}\int \rho(\boldsymbol{r}')\, d\tau'\right] \tag{3}$$

We can write the total charge Q as

$$\boxed{Q = \int \rho(\boldsymbol{r}')\, d\tau'} \tag{4}$$

So the Eq. 3 can be written as

$$\boxed{V_{mono}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}} \tag{5}$$

Here $\rho(\boldsymbol{r}')$ is the charge density function and $\boldsymbol{r}'$ represents the locations where the charges are (which varies over the integral). The net charge Q can also be called the "**monopole moment**".

All of the information about the charge distribution is contained in the calculation for Eq. 4. All of the information about the field point is contained in the calculation for Eq. 5.

At large distances (r), the multipole expansion of a charge distribution is typically dominated by the monopole term. This is because from afar, the distribution appears more and more like a point charge, and the potential can be accurately approximated, as the potential of a pure monopole as given by Coulomb's law for a single point charge,

1.2 The dipole term:

The second term is called the dipole term ($n = 1$)

$$V_{dip}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^2}\int r' \cos\alpha \rho(\boldsymbol{r}')\, d\tau'\right] \tag{6}$$

The dipole term goes like $\frac{1}{r^2}$. If the total charge is zero, the dominant term in the potential will be the dipole term (unless it also vanishes).

We can define the integral (which does not depend on $\boldsymbol{r}$) as the "**dipole moment**" of the distribution.

$$\boxed{\boldsymbol{p} = \int \boldsymbol{r}'\rho(\boldsymbol{r}')\, d\tau'} \tag{7}$$

So the dipole contribution to the potential simplify to

$$\boxed{V_{dip}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{(\boldsymbol{p}\cdot\hat{\boldsymbol{r}})}{r} = \frac{1}{4\pi\epsilon_0}\frac{1}{r}\sum_i \hat{r}_i p_i} \tag{8}$$

Here As usual, $\boldsymbol{r}$ is a unit vector pointing towards the field point P (see Fig. 1). All the information about the charge distribution is contained in the calculation for Eq. 7 and all of the information about the field point is contained in the calculation for Eq. 8.

From Eq. 7 it is clear that the dipole moment is determined by the geometry (size, shape, and density) of charge distribution,

Here we need 3 components (p_x, p_y, p_z) to specify the moment. Each successive term in the multipole expansion involves a larger and larger collection of numbers needed to specify the more-and-more complicated "**moment**" that describes the source, and each successive term falls off by an additional power of r. These moments are called "**tensors**", which can be thought of as an extension of vectors. A "*rank 0*" tensor is a scalar, a "*rank 1*" tensor is a vector, a "*rank 2*" tensor can be represented by a matrix, and so forth.

Name	Symbol (i, j, etc. represent Cartesian coordinates)	Tensor rank	Number of components to specify the moment	r-dependence
Monopole	q	0	1	$1/r$
Dipole	$\mathbf{p}$ or p_i	1	3	$1/r^2$
Quadrupole	$\overleftrightarrow{\mathbf{Q}}$ or Q_{ij}	2	9	$1/r^3$
Octopole	O_{ijk}	3	27	$1/r^4$

Table 1: The table summarizes different terms of "**multipole expansion**" and their decay rate.

1.3 The Quadrupole term:

The next term of Eq. 2 is called the "**quadrupole term**" $(n = 2)$ can be written as

$$V_{\text{quad}}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^3}\int (r')^2\left(\frac{3}{2}\cos^2\alpha - \frac{1}{2}\right)\rho(\boldsymbol{r}')\,d\tau'\right] \tag{9}$$

Here α is the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$. Let's simplify the two terms inside the integral;
The first term:

$$\frac{3}{2}(r')^2\cos^2\alpha = \frac{3}{2}(r'\cos\alpha)^2$$

Now, using index notation, we can write

$$r'\cos\alpha = \hat{\boldsymbol{r}}\cdot\boldsymbol{r}' = \sum_i \widehat{\boldsymbol{r}}_{\boldsymbol{l}}\boldsymbol{r}'_i$$

Where $i, j = x, y, z$. Now the first term can be written as a summation

$$\frac{3}{2}(r'\cos\alpha)^2 = \frac{3}{2}\sum_i\sum_j (\widehat{r}_l r'_i)\left(\widehat{r}_J r'_j\right) \tag{10}$$

The second term:

$$\frac{1}{2}(r')^2 = \frac{1}{2}(r')^2(\hat{\boldsymbol{r}}\cdot\hat{\boldsymbol{r}})$$

$$\hat{\boldsymbol{r}}.\hat{\boldsymbol{r}} = \sum_i \widehat{r}_i\widehat{r}_i$$

Using double summation and Kronecker delta function

$$\hat{\boldsymbol{r}}.\hat{\boldsymbol{r}} = \sum_i\sum_j \widehat{r}_i\widehat{r}_j\delta_{ij}$$

Here $i, j = x, y, z$, the r_i are the components of the vector $\boldsymbol{r}$, and δ_{ij} is the Kronecker's delta function (1 for $i = j$ and 0 for $i \neq j$)

$$\frac{1}{2}\left(r'\right)^2 = \sum_i \sum_j \frac{1}{2}\left(r'\right)^2 \widehat{r}_l \widehat{r}_j \delta_{ij} \tag{11}$$

Piecing together Eq. 9, Eq, 10, and Eq. 11, we get

$$V_{\text{quad}}\left(\boldsymbol{r}\right) = \frac{1}{4\pi\epsilon_0}\left[\frac{1}{r^3}\int\left[\frac{3}{2}\sum_i\sum_j\left(\widehat{r}_l r'_i\right)\left(\widehat{r}_{r_j} r'_j\right) - \sum_i\sum_j\frac{1}{2}\left(r'\right)^2\widehat{r}_l\widehat{r}_j\delta_{ij}\right]\rho\left(\boldsymbol{r}'\right)d\tau'\right]$$

Here we can interchange the order of the integral and the summation

$$\boxed{V_{\text{quad}}\left(\boldsymbol{r}\right) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\sum_i\sum_j\widehat{r}_i\widehat{r}_j\frac{1}{2}\left[\int 3r'_i r'_j - \left(r'\right)^2\delta_{ij}\rho\left(\boldsymbol{r}'\right)d\tau'\right]} \tag{12}$$

Here we have pulled out $\widehat{r}_l\widehat{r}_j$ from each terms and put in front of the integral because the integral is over the primed coordinates.

The integral (stuff in the square bracket) no longer has any $\boldsymbol{r}$ dependence. Thus we have separated out the source information. Equation 12 can now be written as two separate equations,

$$\boxed{Q_{ij} = \int\left(3r'_i r'_j - \left(r'\right)^2\delta_{ij}\right)\rho\left(\boldsymbol{r}'\right)d\tau'} \tag{13}$$

and

$$\boxed{V_{\text{quad}}\left(\boldsymbol{r}\right) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\sum_{ij}\left[\frac{\widehat{r}_l\widehat{r}_j}{2}Q_{ij}\right].} \tag{14}$$

Where $i, j = x, y, z$. Here Q_{ij} is called the "**quadrupole moment**" or the "**quadrupole moment tensor**". Here all of the information about the charge distribution (source) is contained in the calculation for Eq. 13. All of the information about the field point is contained in the calculation for Eq. 14. Once the quadrupole moment tensor is known, the potential can be determined via Eq. 14.

Just like dipole moment $\boldsymbol{p}$ can be written as a 3 -element vector. The quadrupole moment Q is a collection of **nine numbers** (i and j each can be x, y, z) which can be written as a 3×3 matrix.

$$Q_{ij} = \begin{pmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{yx} & Q_{yy} & Q_{yz} \\ Q_{zx} & Q_{zy} & Q_{zz} \end{pmatrix} \tag{15}$$

Moreover, the formula in Eq. 13 is symmetric in reversing i and j, so there are actually *only six independent matrix elements*. The quadrupole moment is a **2-index symmetric tensor**.

1.3.1 Quadrupole moment tensor for continuous charge distribution:

For continuous charge distribution the nine components of Q_{ij} are

$$\boxed{Q_{xx} = \int \left(2x'^2 - y'^2 - z'^2\right) \rho dV'} \tag{16}$$

$$\boxed{Q_{yy} = \int \left(2y'^2 - z'^2 - x'^2\right) \rho dV'} \tag{17}$$

$$\boxed{Q_{zz} = \int \left(2z'^2 - x'^2 - y'^2\right) \rho dV'} \tag{18}$$

Since Q_{ij} is symmetric

$$\boxed{Q_{xy} = Q_{xy} = \int (3x'y') \rho dV'} \tag{19}$$

$$\boxed{Q_{yz} = Q_{zy} = \int (3y'z') \rho dV'} \tag{20}$$

$$\boxed{Q_{zx} = Q_{xz} = \int (3z'x') \rho dV'} \tag{21}$$

Also note that Q_{ij} is traceless,i.e.

$$\boxed{Q_{xx} + Q_{yy} + Q_{zz} = 0} \tag{22}$$

Therefore Q only has 5 independent components.

1.3.2 Quadrupole moment tensor for discrete point charge distribution:

For discrete distribution of n point charges q_i at positions $\mathbf{r_i}'$ the nine components of Q_{ij} are

$$\boxed{Q_{xx} = \sum_{i=1}^{n} q_i \left(2x_i'^2 - y_i'^2 - z_i'^2\right)} \tag{23}$$

$$\boxed{Q_{yy} = \sum_{i=1}^{n} q_i \left(2y_i'^2 - z_i'^2 - x_i'^2\right)} \tag{24}$$

$$\boxed{Q_{zz} = \sum_{i=1}^{n} q_i \left(2z_i'^2 - x_i'^2 - y_i'^2\right)} \tag{25}$$

$$\boxed{Q_{xy} = Q_{xy} = \sum_{i=1}^{n} q_i \left(3x_i'y_i'\right)} \tag{26}$$

$$\boxed{Q_{yz} = Q_{zy} = \sum_{i=1}^{n} q_i \left(3y_i'z_i'\right)} \tag{27}$$

$$\boxed{Q_{zx} = Q_{xz} = \sum_{i=1}^{n} q_i \left(3 z_i' x_i'\right)} \tag{28}$$

Example-1: Simple Quadrupole

Question: Find the quadrupole moment tensors of a simple quadrupole-for alternating charges $\pm Q$ at the corners of a square (hence the name quadrupole) as shown in Fig. 2.

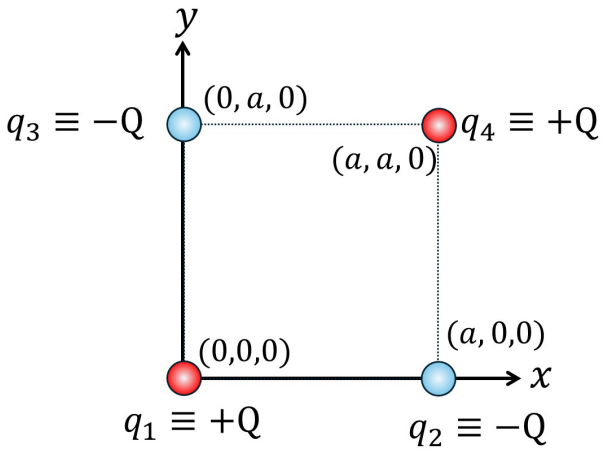


Figure 2: Simple quadrupole of alternating point charges $\pm Q$ at the corners of a square

Solution: It is easy to see that this 4-charge system has zero net charge as well as zero net dipole moment, so the leading term in its potential at long distances is the quadrupole potential .

Here assume

$$q_1 \equiv +Q@(0,0,0)$$
$$q_2 \equiv -Q@(a,0,0)$$
$$q_3 \equiv -Q@(0,a,0)$$
$$q_4 \equiv +Q@(a,a,0)$$

Following the definition of the quadrupole moments for point charges let's calculate

$$Q_{xx} = \sum_{i=1}^{4} q_i \left(2x_i'^2 - y_i'^2 - z_i'^2\right)$$
$$= q_1 \left(2x_1'^2 - y_1'^2 - z_1'^2\right) + q_2 \left(2x_2'^2 - y_2'^2 - z_2'^2\right) + q_3 \left(2x_3'^2 - y_3'^2 - z_3'^2\right) + q_4 \left(2x_4'^2 - y_4'^2 - z_4'^2\right)$$
$$= Q\left(2\times 0^2 - 0^2 - 0^2\right) - Q\left(2\times a^2 - 0^2 - 0^2\right) - Q\left(2\times 0^2 - a^2 - 0^2\right) + Q\left(2\times a^2 - a^2 - 0^2\right)$$
$$Q_{xx} = -Q\left(2a^2\right) + Q\left(a^2\right) + Q\left(a^2\right) = 0$$

Similarly

$$Q_{yy} = \sum_{i=1}^{4} q_i \left(2y_i'^2 - z_i'^2 - x_i'^2\right)$$
$$= q_1 \left(2y_1'^2 - z_1'^2 - x_1'^2\right) + q_2 \left(2y_2'^2 - z_2'^2 - x_2'^2\right) + q_3 \left(2y_3'^2 - z_3'^2 - x_3'^2\right) + q_4 \left(2y_4'^2 - z_4'^2 - x_4'^2\right)$$
$$= Q\left(2\times 0^2 - 0^2 - 0^2\right) - Q\left(2\times 0^2 - 0^2 - a^2\right) - Q\left(2\times a^2 - 0^2 - 0^2\right) + Q\left(2\times a^2 - a^2 - 0^2\right)$$
$$Q_{yy} = Q\left(a^2\right) - Q\left(2a^2\right) + Q\left(a^2\right) = 0$$

Also

$$Q_{zz} = \sum_{i=1}^{4} q_i \left(2z_i'^2 - x_i'^2 - y_i'^2\right)$$
$$= q_1 \left(2z_1'^2 - x_1'^2 - y_1'^2\right) + q_2 \left(2z_2'^2 - x_2'^2 - y_2'^2\right) + q_3 \left(2z_3'^2 - x_3'^2 - y_3'^2\right) + q_4 \left(2z_4'^2 - x_4'^2 - y_4'^2\right)$$
$$= Q\left(2\times 0^2 - 0^2 - 0^2\right) - Q\left(2\times 0^2 - a^2 - 0^2\right) - Q\left(2\times 0^2 - 0^2 - a^2\right) + Q\left(2\times 0^2 - a^2 - a^2\right)$$
$$Q_{zz} = Q\left(a^2\right) + Q\left(a^2\right) - Q\left(2a^2\right) = 0$$

Therefore all the diagonal moments vanish. Lets calculate the off-diagonal moments

$$Q_{xy} = Q_{xy} = \sum_{i=1}^{4} q_i \left(3x_i' y_i'\right)$$
$$= q_1 \left(3x_1' y_1'\right) + q_2 \left(3x_2' y_2'\right) + q_3 \left(3x_3' y_3'\right) + q_4 \left(3x_4' y_4'\right)$$
$$= Q(3\times 0\times 0) - Q(3\times a\times 0) - Q(3\times 0\times a) + Q(3\times a\times a) = 3a^2 Q$$
$$Q_{yz} = Q_{zy} = \sum_{i=1}^{n} q_i \left(3y_i' z_i'\right)$$
$$= q_1 \left(3y_1' z_1'\right) + q_2 \left(3y_2' z_2'\right) + q_3 \left(3y_3' z_3'\right) + q_4 \left(3y_4' z_4'\right)$$
$$= Q(3\times 0\times 0) - Q(3\times 0\times 0) - Q(3\times a\times 0) + Q(3\times a\times 0) = 0$$
$$Q_{zx} = Q_{xz} = \sum_{i=1}^{n} q_i \left(3z_i' x_i'\right)$$
$$= q_1 \left(3z_1' x_1'\right) + q_2 \left(3z_2' x_2'\right) + q_3 \left(3z_3' x_3'\right) + q_4 \left(3z_4' x_4'\right)$$
$$= Q(3\times 0\times 0) - Q(3\times 0\times a) - Q(3\times 0\times 0) + Q(3\times 0\times a) = 0$$

The quadrupole tensor in matrix from is given by

$$Q_{ij} = \begin{pmatrix} Q_{xx} & Q_{xy} & Q_{xz} \\ Q_{yx} & Q_{yy} & Q_{yz} \\ Q_{zx} & Q_{zy} & Q_{zz} \end{pmatrix}$$

Finally we can conclude that the quadrupole tensor of an simple quadrupole is given by

$$Q_{ij} = \begin{pmatrix} 0 & 3a^2 Q & 0 \\ 3a^2 Q & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Note: One can also use Eq. 16−Eq. 21 to calculate quadrupole tensor with charge density given by

$$\rho(\boldsymbol{r}) = Q\delta^3(\boldsymbol{r}) - Q\delta^3(\boldsymbol{r} - a\widehat{\boldsymbol{x}}) - Q\delta^3(\boldsymbol{r} - a\widehat{\boldsymbol{y}}) + Q\delta^3(\boldsymbol{r} - a\widehat{\boldsymbol{x}} - a\widehat{\boldsymbol{y}})$$

The three dimensional Dirac delta functions in Cartesian coordinates become, for example

$$\delta^3(\boldsymbol{r} - a\widehat{\boldsymbol{x}} - a\widehat{\boldsymbol{y}}) = \delta(z)\delta(x-a)\delta(y-a)$$

and so on. In the limit $a \to 0$, we can define a "*pure quadrupole moment tensor*" in the same way as we built a dipole moment vector.

Example-2: Homogeneously Charged Ellipsoid

Question: Consider a homogeneously charged ellipsoid $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 \leq 1$ and calculate the quadrupole tensor Q_{ij} with respect to the ellipsoids center. Check to make sure that the result for Q_{ij} is traceless. ~~Find Q_{ij} for the special case $a = b$.~~

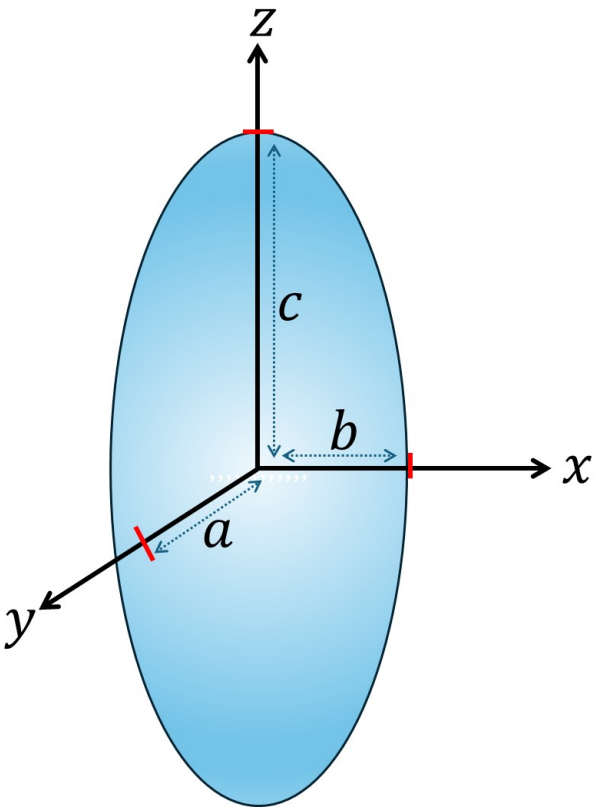


Figure 3: Homogeneously charged ellipsoid with charge density ρ_0

Solution: Lets consider principal axes of the homogeneously charged ellipsoid coincide with the coordinate axes (as shown in Fig. 3). A suitable coordinate system for this problem will be "**ellipsoidal coordinates**" given by

$$\begin{aligned} x &= as\sin\theta\cos\varphi \\ y &= bs\sin\theta\sin\varphi \\ z &= cs\cos\theta \end{aligned}$$

In this coordinate system the infinitesimal volume element is given by

$$dV = dxdydz = \left(abcs^2\sin\theta\right) dsd\theta d\varphi$$

Now we can evaluate the quadrupole moments using the Eq. 16−Eq. 21. It is not necessary to evaluate all quadrupole moments.

We just have to calculate the Q_{xx} and Q_{yy} moments and find the Q_{zz} moment from the traceless property of quadrupole tensor ($Q_{xx} + Q_{yy} + Q_{zz} = 0$). Following the definition

of the quadrupole moments as in Eq. 16, let's calculate

$$Q_{xx} = \int \left(2x'^2 - y'^2 - z'^2\right) \rho dV'$$

$$= \int \left(2a^2 s^2 \sin^2\theta \cos^2\varphi - b^2 s^2 \sin^2\theta \sin^2\varphi - c^2 s^2 \cos^2\varphi\right) \rho_0 \left(abcs^2 \sin\theta\right) ds d\theta d\varphi$$

$$= \rho_0(abc) \int_0^1 s^4 ds \int_0^{2\pi} \int_0^{\pi} \left(2a^2 \sin^2\theta \cos^2\varphi - b^2 \sin^2\theta \sin^2\varphi - c^2 \cos^2\theta\right) (\sin\theta) d\theta d\varphi$$

let us first integrate w.r.t s first

$$= \rho_0(abc)\frac{1}{5} \int_0^{2\pi} \left[2a^2 \cos^2\varphi \int_0^{\pi} \sin^3\theta d\theta - b^2 \sin^2\varphi \int_0^{\pi} \sin^3\theta - c^2 \cos^2\varphi \int_0^{\pi} \sin\theta d\theta\right] d\varphi$$

Then integrating over θ and φ we obtain

$$Q_{xx} = \frac{4\pi}{15}\rho_0 abc \left[2a^2 - b^2 - c^2\right]$$

Similarly

$$Q_{yy} = \int \left(2y'^2 - z'^2 - x'^2\right) \rho dV'$$

$$= \rho_0(abc) \int_0^1 s^4 ds \int_0^{2\pi} \int_0^{\pi} \left(2b^2 \sin^2\theta \sin^2\varphi - a^2 \sin^2\theta \cos^2\varphi - c^2 \cos^2\varphi\right) (\sin\theta) d\theta d\varphi$$

$$= \frac{4\pi}{15}\rho_0 abc \left[2b^2 - a^2 - c^2\right]$$

Since the quadrupole tensor is traceless

$$Q_{zz} = -Q_{xx} - Q_{yy}$$

$$Q_{zz} = \frac{4\pi}{15}\rho_0 abc \left[2c^2 - a^2 - b^2\right]$$

The off-diagonal moments should all vanish since we are already in a coordinate system where the axes of the ellipsoid are coinciding with the coordinate axes. Let us see if this is true at the example of the Q_{xy} moment given by Eq. 19:

$$Q_{xy} = Q_{xy} = \int (3x'y') \rho dV'$$

$$= \rho_0(abc) \int_0^1 s^4 ds \int_0^{2\pi} \int_0^{\pi} (\sin\theta \cos\varphi \sin\theta \sin\varphi)(\sin\theta) d\theta d\varphi$$

$$= \rho_0(abc) \left(\frac{1}{5}\right) \int_0^{2\pi} \int_0^{\pi} \left(\sin^3\theta\right) (\cos\varphi \sin\varphi) d\theta d\varphi$$

$$= \rho_0(abc) \left(\frac{1}{5}\right) \int_0^{\pi} \sin^3\theta d\theta \int_0^{2\pi} \frac{1}{2} \sin 2\varphi d\varphi = 0$$

$$Q_{xy} = Q_{xy} = 0$$

Finally we can conclude that the quadrupole tensor of an ellipsoid with a homogeneous mass distribution is given by

$$Q_{ij} = \frac{q}{5} \begin{pmatrix} 2a^2 - b^2 - c^2 & 0 & 0 \\ 0 & 2b^2 - a^2 - c^2 & 0 \\ 0 & 0 & 2c^2 - a^2 - b^2 \end{pmatrix}$$

Here the total charge of the ellipsoid is $q = \frac{4\pi}{3} abc\rho_0$.
For the case when, $a = b$ we have

$$Q_{ij} = \frac{q}{5} \begin{pmatrix} a^2 - c^2 & 0 & 0 \\ 0 & a^2 - c^2 & 0 \\ 0 & 0 & 2\left(c^2 - a^2\right) \end{pmatrix}$$

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